Software

No Pain, No Gain - A Software Playbook on the SaaS Transition

28 February 2022

Key Takeaway

The software industry is going through a fundamental business model switch from traditional licences to SaaS-based subscription. Our analysis of previous transitions among 4 studied companies suggests the journey is ultimately positive, as it increases sales growth by 6ppts, margins by 250bps and EV/ sales valuations by c3 multiple points. However, the journey is far from linear and typically costs c1.5 yrs of profits.

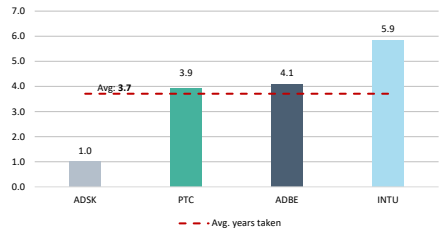
A fundamental shift: The shift to SaaS is an inescapable global trend that is impacting the global software sector. Every software company in Europe is at different stages through this transition and, as a result, we expect it to remain the dominant theme impacting companies for the next 2-3 years. This note analyses the transition impacts of 4 companies that have been through the SaaS transition and 4 companies that are currently midway through the transition. The overall conclusion is that the SaaS transition is ultimately positive, but the journey is far from linear.

The gain: The shift to SaaS increases flexibility for customers, while at the same time making software easier for customers to use and consume. This has a direct positive impact on operational metrics such as customer numbers and customer satisfaction. In turn, this drives very favourable impacts on the financials. On average, our analysis suggests that the transition accelerates revenue growth by avg. 6%, improves margins by avg. 250bps. It also improves the quality of earnings by increasing the proportion of recurring revenues.

The pain: While the SaaS transition is ultimately positive, the most immediate impact for investors is a painful "J-curve". The switch from up front revenue recognition to more ratable revenue recognition, pushes revenues to the right. On average, this negatively impacts near term revenue growth by c10%. As revenue will eventually be recognised and future revenue growth will accelerate, management teams rarely adjust cost bases to match a short term dip in revenue growth. As a result, slower revenues in successful transitions typically lead to c15% margin compression.

Impacts on valuation: Incurring (often material) near-term downgrades, in order to drive mid-term benefits is a difficult message for investors. While everyone knows the long-term benefits outweigh the near term costs, the average share price move on the announcement of a transition is -9%. On a medium term view, we note all 4 studied US companies that completed transitions have seen a positive valuation re-rating, with EV/ sales valuations rising by c3 multiple points. However, we have yet to see any evidence in Europe that a SaaS transition is beneficial for valuation. For instance, Sage, who is now near end of the transition, is trading at 40% discount vs European software peers. This compares to historical average of 18% discount. Notwithstanding being c9 years into the transition, SAP is trading at 30% discount compared to peers, vs trading in-line historically.

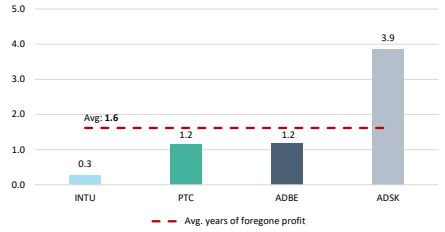
Chart 1 - Length of transition to cloud varies among studied companies



Source: Jefferies, Company data

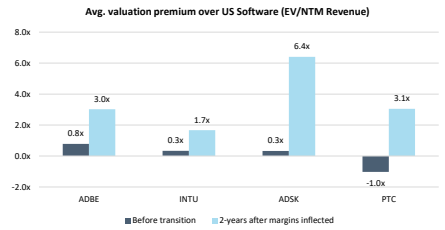
Chart 2 - ...as well as cost of transition

Foregone profit = Adj. EBIT



Source: Jefferies, Company data

Chart 3 - Coming out of the transitions, companies see multiple expansion



Source: Jefferies, Company data, FactSet

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^Prior trading day closing price unless otherwise noted.

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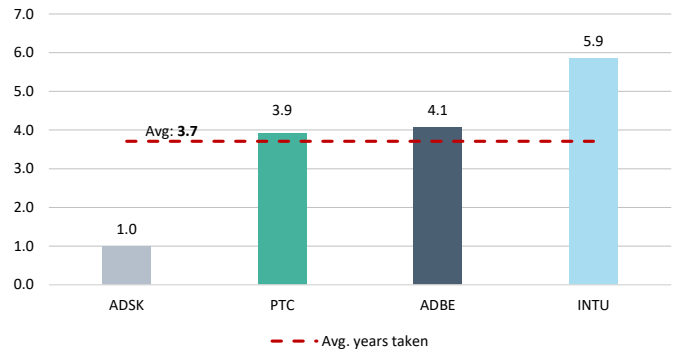
Please see analyst certifications, important disclosure information, and information regarding the status of non-US analysts on pages 12 to 18 of this report.

* Jefferies International Limited

Key charts

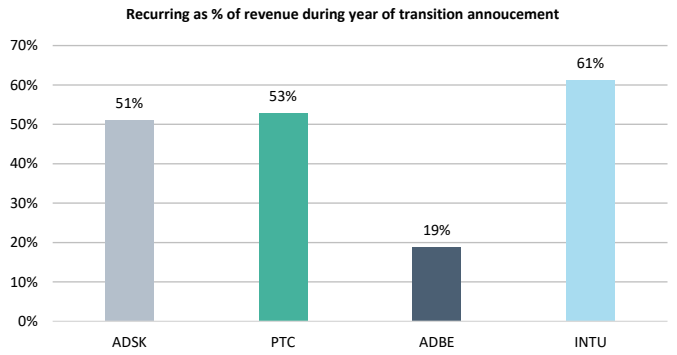
We believe the transition is ultimately positive for software companies.

Chart 4 - The duration of SaaS transitions varies among studied companies



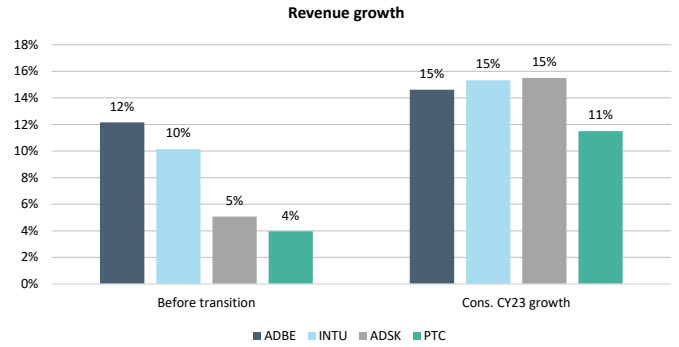
Source: Jefferies, Company data

Chart 5 - ... given different product complexity, different customer bases, and different starting points in the journey



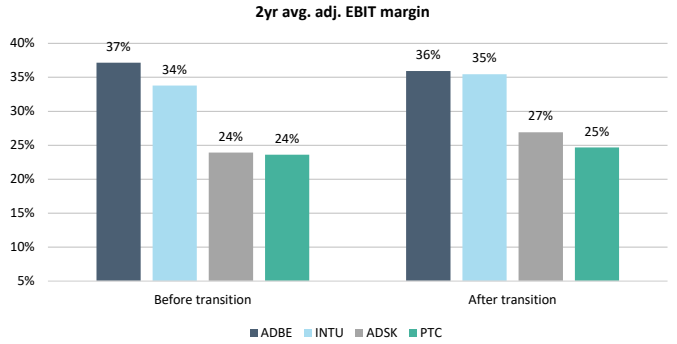
Source: Jefferies, Company data

Chart 6 - Upon completion of the transition, companies are forecasted to deliver noticeably faster revenue growth



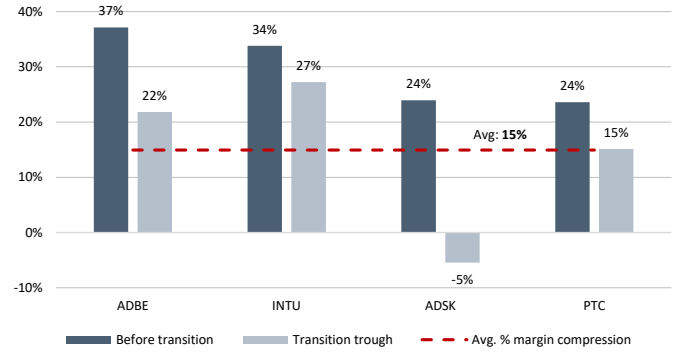
Source: Jefferies, Company data

Chart 7 - In addition, margins ultimately rise



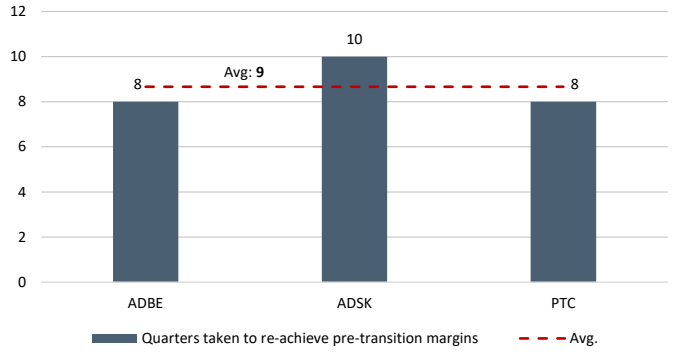
Source: Jefferies, Company data

Chart 8 - However, the gains come with costs, as companies often go through a period of margin compression



Source: Jefferies, Company data

Chart 9 - It took an average 8 quarters for margins to return to pre-transition level among studied companies



Source: Jefferies, Company data

The SaaS transition

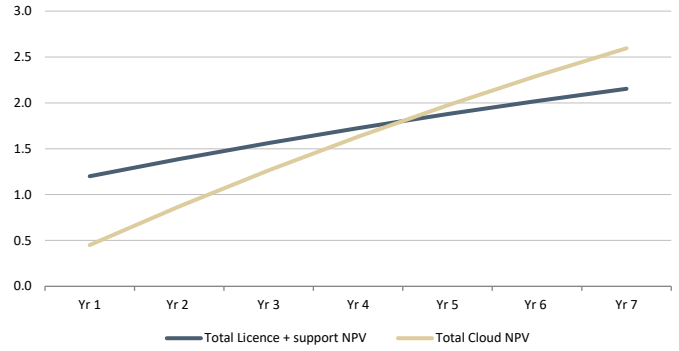
A SaaS transition is characterized by 2 meaningful changes. Firstly, from an architectural point of view, the software moves from being implemented "on-premise" in customer systems and managed by the customer to being hosted in a cloud-environment (either private or public) and managed by the software vendor. Secondly, the contracting structure changes, moving from a licence (a right to use the software) + maintenance (typically 20% of the licence, paid annually) model to a subscription model, where the licence and maintenance is bundled into a single price and recognized ratably through the contract.

We think the architectural shift drives material operational "gains", that ultimately leads to accelerating financials. However, the contracting shift defers revenue recognition, and creates a J-curve profile, with near P&L "pains", before the medium term benefits kick in.

Assessing these up front costs versus medium term benefits is a material challenge for investors and the subject of this note.

Chart 10 - Transition to the subscription model ultimately delivers higher NPV compared to traditional licence model.

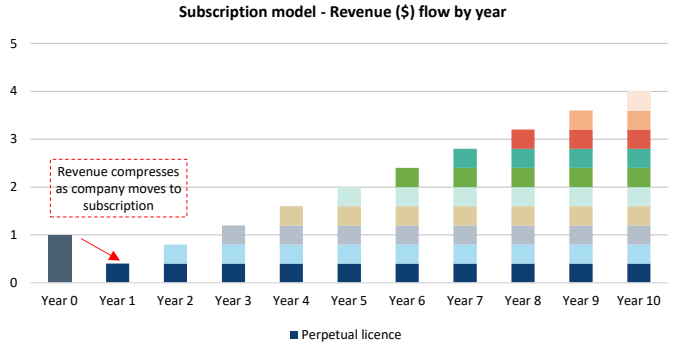
$1 licence contract with 20% of contract value being maintenance vs subscription contract recognizing $0.45 annually



Source: Jefferies

Chart 11 - The incremental benefits of recurring revenue accelerate into the future

Assume flat new booking of $1 every year with no churn and no price increase



Source: Jefferies

The gain(s)

We believe there are a number of operational advantages in embracing a SaaS transition. We list 5 key advantages below:

1. **Increased scalability** - cloud-based infrastructure scales to support fluctuating workloads. This is a more efficient than buying infrastructure to accommodate peak demand, leading to sub-optimal utilization in off-peak periods.
2. **Supporting greater flexibility** - cloud-based computing allows you to access data easier from outside the organisation. This supports greater business flexibility whether that is empowering mobile workers, supporting working from home or facilitating improved collaboration and data sharing.
3. **Automatic updates** - cloud-based software is managed by the software company and not the customers, meaning that all customers can be kept on the same version of software. This makes it easier for the software company to develop new functionality and easier for customers to consume this new innovation.

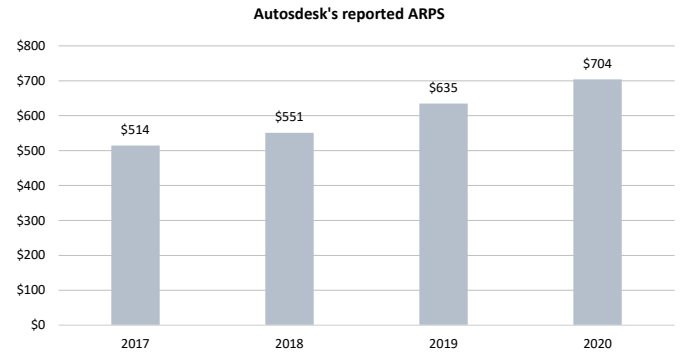
4. **Cross-selling** - Seamless cloud-based software should create an improved customer experience. This makes it more likely that customers will buy additional functionality and cloud-based delivery makes this an effortless process. This provides a platform for enhanced cross-selling and higher net retention.
5. **Capturing infrastructure spend** - Software companies typically charge for a holistic solution, which enables them to capture the infrastructure cost of hosting the solution. Software companies can leverage the hyperscalers, so this does not require any additional capex and provides low-risk additional revenue.

These operational advantages suggest there should be clear benefits from moving from traditional licences to a SaaS based model. Encouragingly, we think these show through in operational metrics.

- **Improvement in customer satisfaction.** The shift to subscription starts to harmonise customers onto the latest versions of software, that are managed by the software company rather than the customer. This typically leads to improving customer satisfaction, for instance Sage recorded 4 consecutive years of improvement in Net Promoter Score (NPS), from 3 reported in FY14 to 31 reported in FY18. Or more recent example would be SAP, where NPS has improved materially for 2 consecutive years (-6 in 2019 to 4 in 2020 and 10 in 2021).
- **Higher revenues per customer.** A cloud based platform provides an easier, fully integrated way for customers to buy and use new functionality. Together with improved customer satisfaction, which makes customers happier to buy more, this is a vehicle to drive higher revenues per customer, see Chart 12.
- **Increasing customer numbers.** The ease of consumption, together with the lower up front costs acts as an accelerator to customer growth, see Chart 13, 14 and 15.

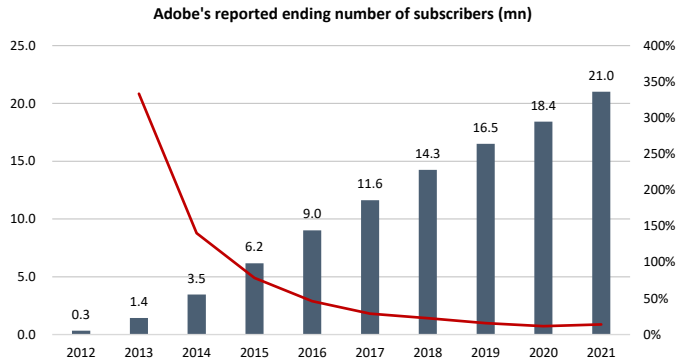
Chart 12 - Autodesk reported improving annualized revenue per subscription

shown for FY



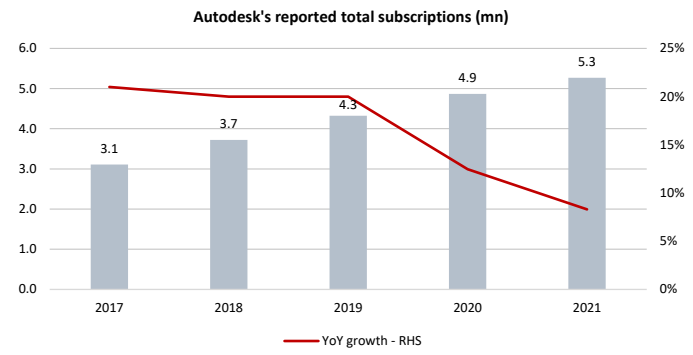
Source: Jefferies, Company data

Chart 13 - By moving to the subscription model, Adobe has been able to expand its customer base at a rapid pace



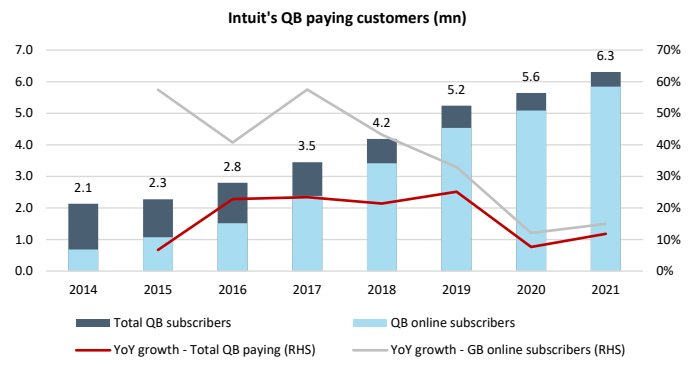
Source: Jefferies, Company data

Chart 14 - Subscriptions have also accelerated at Autodesk



Source: Jefferies, Company data

Chart 15 - ...and Intuit, where online QuickBooks (QB) subscribers contribute to majority of expansion in paying Small Business subscribers



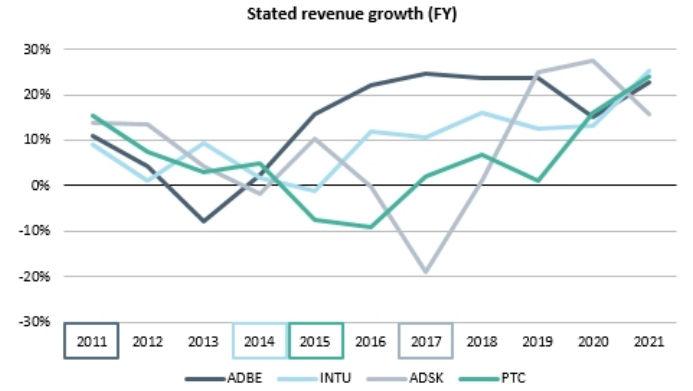
Source: Jefferies, Company data

As operational metrics improve, this ultimately drives accelerating financials. To begin with revenue growth weakens as licence revenues give way to more ratable revenue recognition. However, once the share of licence sales shrinks to a point where it no longer negatively impacts group revenues, the incremental benefits of subscriptions kick in. This is clearly visible in Chart 16 that shows accelerating revenue growth. Meanwhile, Chart 17 shows that margins follow revenue momentum, initially falling and then re-accelerating.

In terms of magnitude, we estimate that a SaaS transition accelerates growth by c6 ppts and margins by 250bps compared to before the transition.

Chart 16 - All studied companies reported accelerated revenue growth after transition

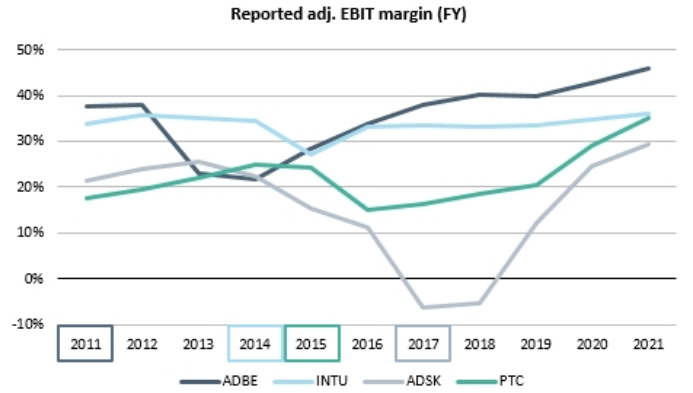
Rectangular boxes highlight year of announced transition



Source: Jefferies, Company data

Chart 17 - ...and better margins

Rectangular boxes highlight year of announced transition

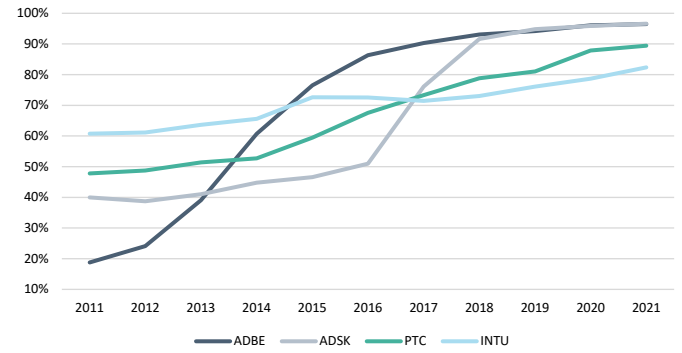


Source: Jefferies, Company data

Importantly, not only do revenue growth and margins improve, but the quality of earnings also improves. This is visible in recurring revenues increasing as a proportion of revenues, see Chart 18 and seasonality improving, see Intuit in Chart 19.

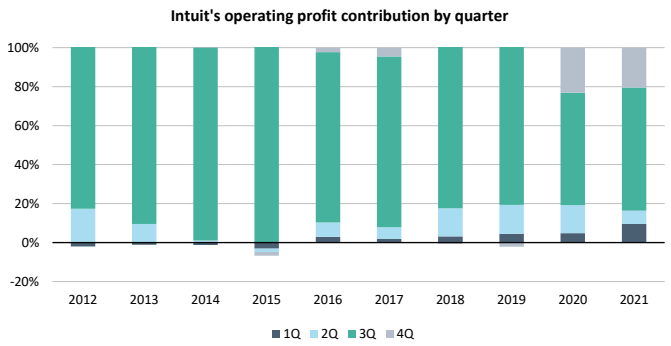
Chart 18 - Recurring revenues increase as a proportion of the overall revenue mix

FY shown



Source: Jefferies, Company data

Chart 19 - Accelerated cloud transition helps reduce P&L seasonality for Intuit (usually heavily 3Q loaded given tax returns season in the US)



Source: Jefferies, Company data

The cost(s)

We think the benefits of a SaaS transition are clear, with accelerating growth, improving margins and quality of earnings. However, the transition is not cost free. More ratable revenue recognition defers revenue recognition. Furthermore, we observe companies increased core operating expenses by average of 5%, in order to (1) invest in subscription or cloud-based technology components that would allow for more seamless users experience and (2), educate and remunerate sales on the new GTM strategy and ensure momentum going forward, thus putting pressure on near term revenues and margins.

Every transition is different, which makes generalizations difficult. The encouraging news, is that costs are temporary. On average, it takes around 6 quarters for margins to trough and 9 quarters for margins to return to pre-transition revenues.

If we simplistically take this window of depressed margins as a view of the transition costs, then we estimate the average transition costs c1.5 years of profit.

Exhibit 1 - Margin compression adjusted for transition duration

	Announced quarter	Announced qtr. adj. EBIT %	Trough margin quarter	Reported trough margin	Qtrs taken to reach trough	Avg. compression per quarter
ADBE	4Q11	39%	3Q14	19%	7	2.8%
ADSK	1Q16	15%	4Q17	-17%	7	4.5%
PTC	1Q16	21%	4Q16	11%	3	3.3%
INTU	2013	35%	2015	18%	8	2.1%

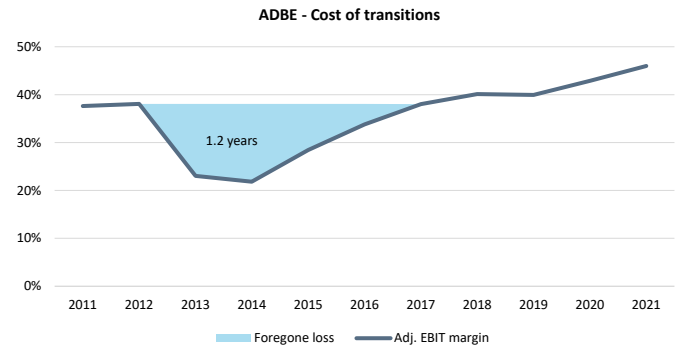
Source: Jefferies, Company data

Exhibit 2 - We estimate the cost of transition for Adobe to be 1.2 years of profit

Estimated cost of transition - Adobe		
Pre-transition revenue	$mn	4,310
Pre-transition adj. EBIT	$mn	1,632
Pre-transition margin		38%
Transition period:		
Year 1 margin		23%
Margin compression		15%
Year 1 foregone profit	$mn	638
Year 2 margin		22%
Year 2 foregone profit	$mn	691
Year 3 margin		28%
Year 3 foregone profit	$mn	406
Year 4 margin		34%
Year 4 foregone profit	$mn	175
Total foregone profit	$mn	1,910
Years of foregone profit	yrs	1.2

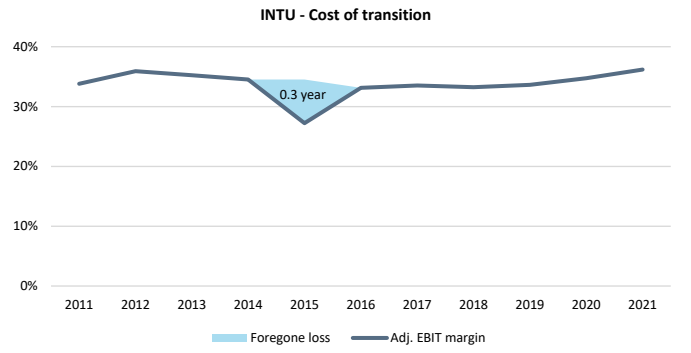
Source: Jefferies, Company data

Chart 20 - 1.2 years of foregone profit estimated for Adobe



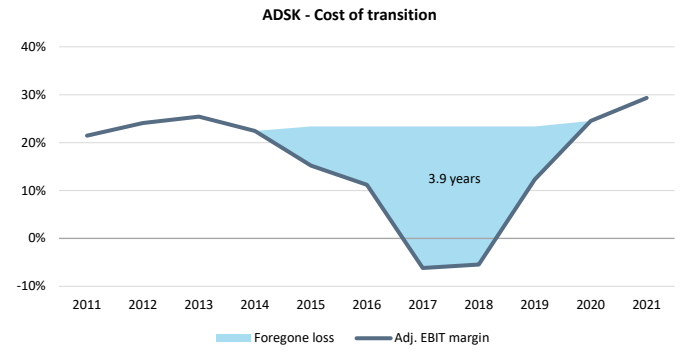
Source: Jefferies, Company data

Chart 21 - For Intuit, our estimated cost is 0.3 year



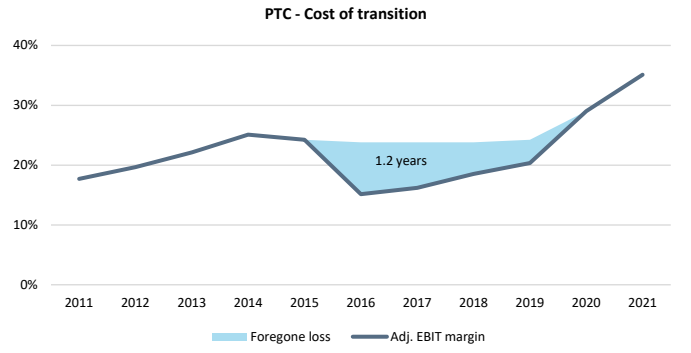
Source: Jeffferies, Company data

Chart 22 - The cost for Autodesk is higher at approximately 4 years



Source: Jefferies, Company data

Chart 23 - For PTC, the cost is approximately 1.2 years



Source: Jefferies, Company data

Ongoing transitioners - A few highlights

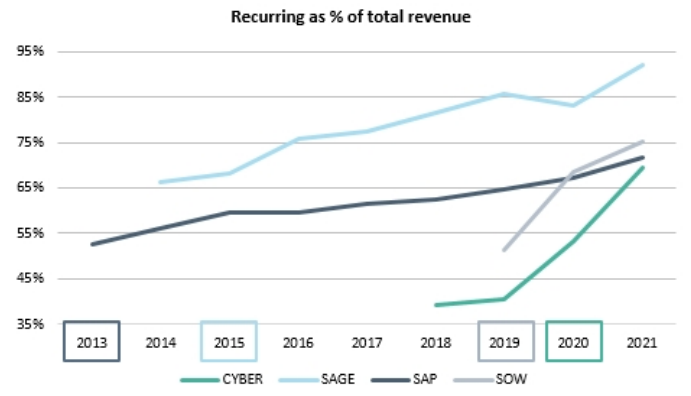
The sample size of companies that have been through the SaaS transition remains small and led by US-based companies. In Europe, companies are still largely in the midst of the transition. As a result, we also look at the following transitioners; SAP, Sage and Software AG in Europe and CyberArk in the US.

Among the transitioners in our study, SAP has been in the process of migrating customers to cloud offerings and subscription model the longest - 9 years. It is then followed by Sage, who announced their plan in 2015, Software AG in 2019 and CyberArk in 2020.

Given product complexity and the mission critical nature of the software, SAP has net been able to force the pace of a SaaS transition. This has resulted in an extended transition, with modest margin compression. It has arguably still cost 1 year of profits, but this impact has been spread over a longer time frame.

With regard to Sage, Software AG and CyberArk, management teams have recently indicated they are now reaching the end of the transition where margins have troughed. When comparing most recently reported share of recurring vs of those of companies that have completed the transition, we notice c75-80% of recurring revenue is where margins start inflecting. At this point, the portion of license revenue becomes small enough to not make a big swing in revenue and drag down margins.

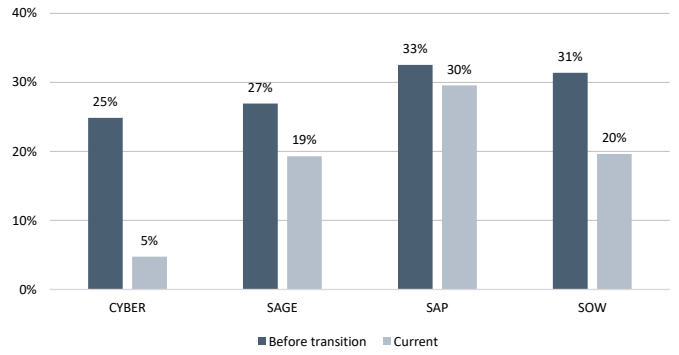
Chart 24 - The pace of transition varies across the sector



Source: Jefferies, Company data

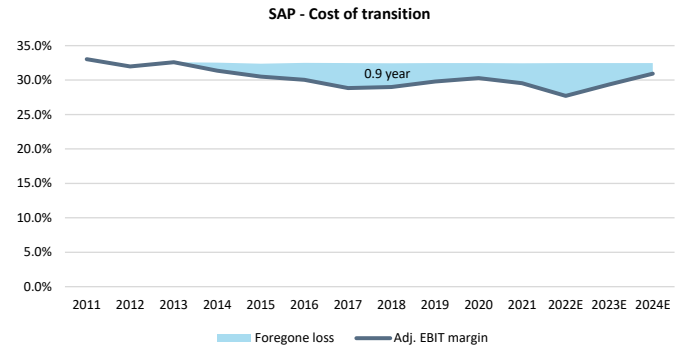
Chart 25 - The faster the transition, the more severe the margin impact

*Current shown for most recent reported numbers



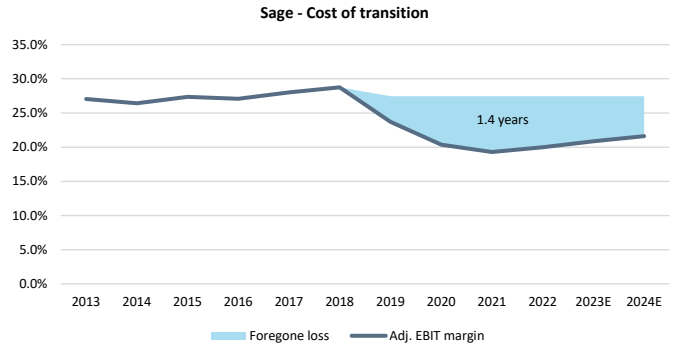
Source: Jefferies, Company data

Chart 26 - We estimate SAP will forego nearly 1 year of profit by end of FY2024



Source: Jefferies, Company data

Chart 27 - For Sage, the cost is higher at 1.4 years

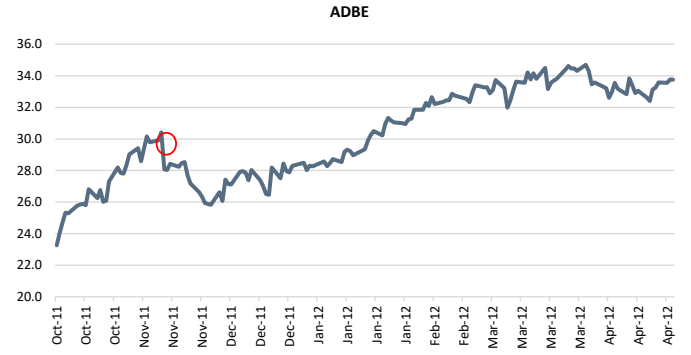


Source: Jefferies, Company data

Tying to valuation

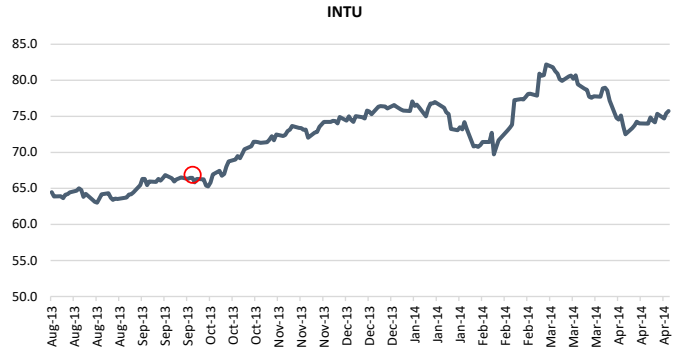
We take a look at share price reaction on the day of companies announcing or indicating an acceleration in the transition. On average, the share price dropped 9% due to a combination of P&L downgrades and greater uncertainty.

Chart 28 - Adobe stock dropped 8% on the day announcing transition



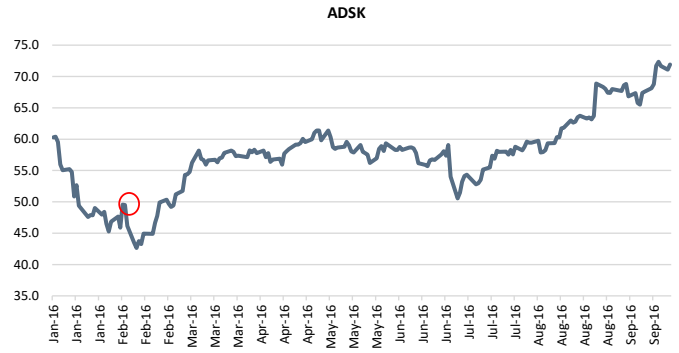
Source: Jefferies, Company data, FactSet

Chart 29 - Meanwhile, Intuit share remained relatively flat following the announcement, declining 1%



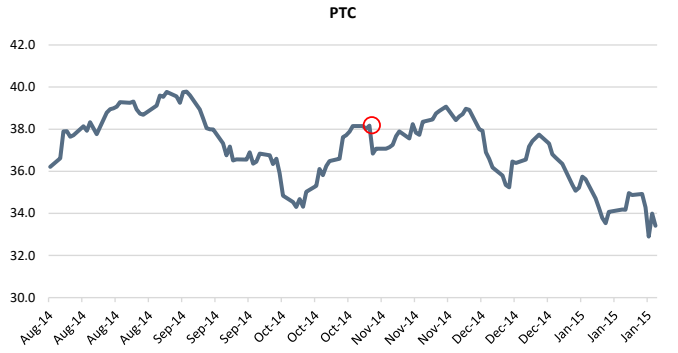
Source: Jefferies, Company data, FactSet

Chart 30 - Autodesk share dropped 14% the days following announcement of moving business to the cloud



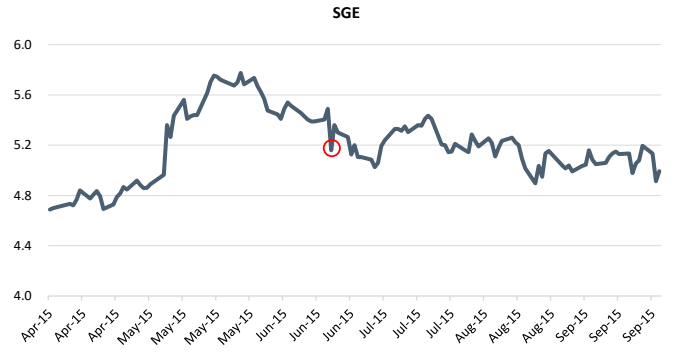
Source: Jefferies, Company data, FactSet

Chart 31 - PTC share price dropped 3% following the day it announced acceleration into subscription model



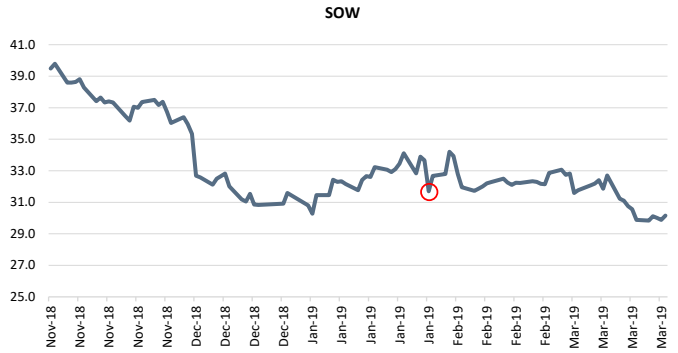
Source: Jefferies, Company data, FactSet

Chart 32 - Among the transition incumbents, Sage dropped 6% on the day of the announcement



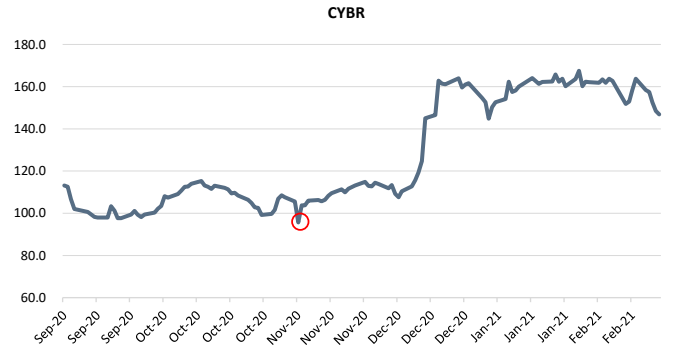
Source: Jefferies, Company data, FactSet

Chart 33 - Similarly, Software AG dropped 6% as it downgraded revenue growth and guided to margins compression



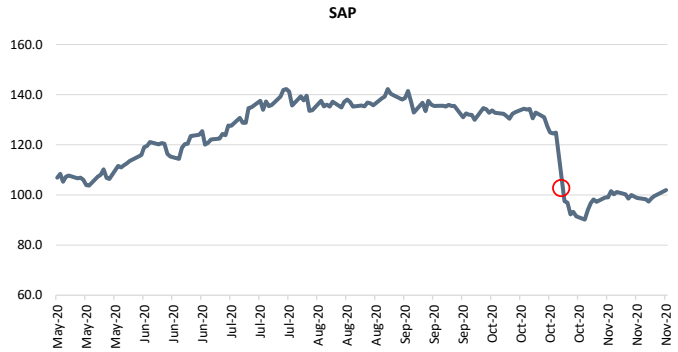
Source: Jefferies, Company data, FactSet

Chart 34 - CyberArk dropped 9% after reporting transition headwinds impact and plan to accelerate to the subscription model



Source: Jefferies, Company data, FactSet

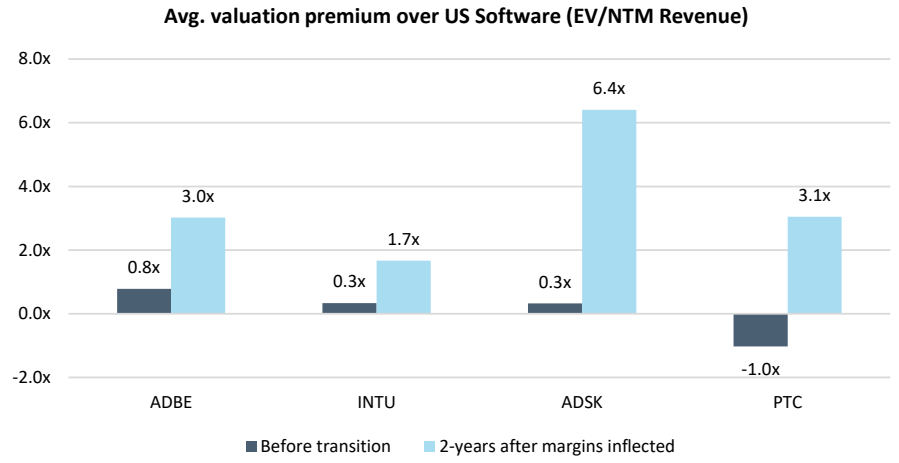
Chart 35 - SAP's shares dropped 22% on news of a faster shift to cloud and updated guidance. Shares continued falling in the aftermath



Source: Jefferies, Company data, FactSet

But there is light at the end of the tunnel. Once companies are through the transition, valuations typically improve. Our 4 studied companies saw EV/ sales valuations rise by c3 multiple points.

Chart 36 - Valuation upgraded following inflection of margins and better revenue growth

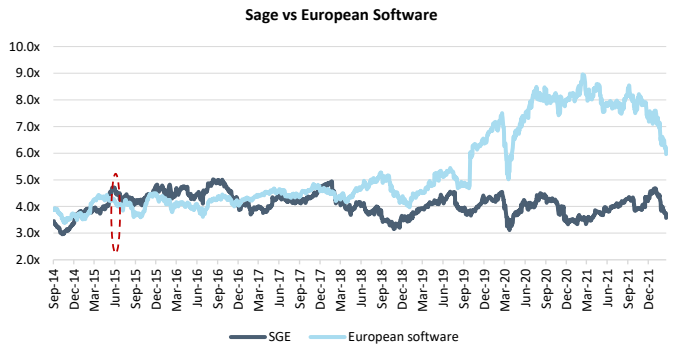


Source: Jefferies, Company data

On the names that are still undergoing the transitions, there is still no evidence of any valuation upside from the transition. As seen in the charts below, there is no obvious valuation benefit at Sage, SAP or Software AG.

Chart 37 - Going through the model transition, Sage has largely underperformed the sector

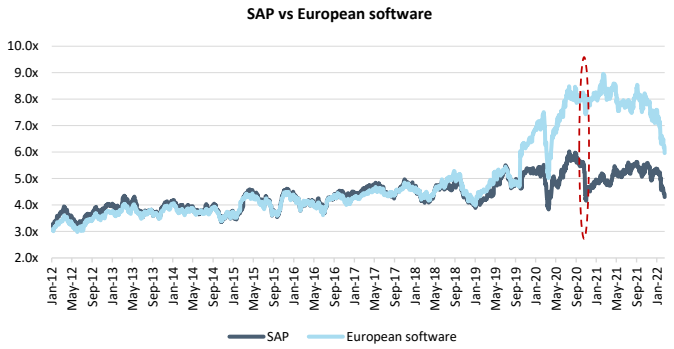
NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Chart 38 - Similarly, there is no valuation upside at SAP

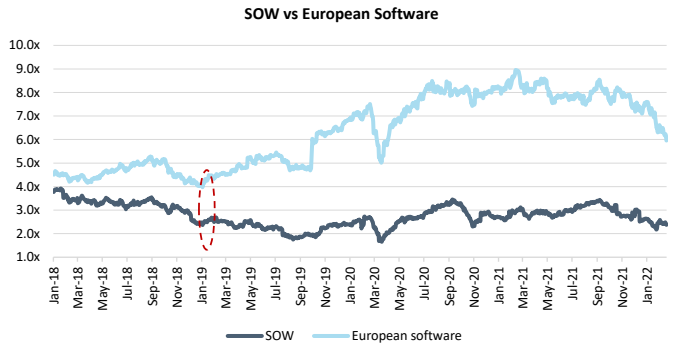
NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Chart 39 - Software AG has de-rated versus the sector

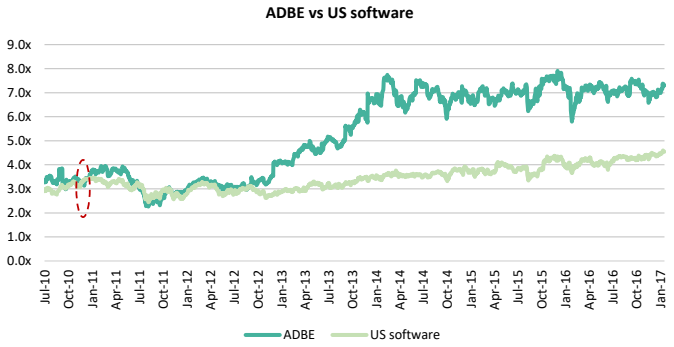
NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Chart 40 - Meanwhile, Adobe started trading at premium over US software peers c2 years into the transition

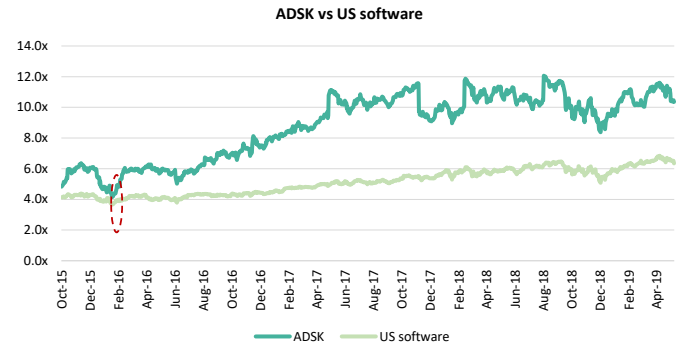
NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Chart 41 - Looking at Autodesk, the timeline is even shorter

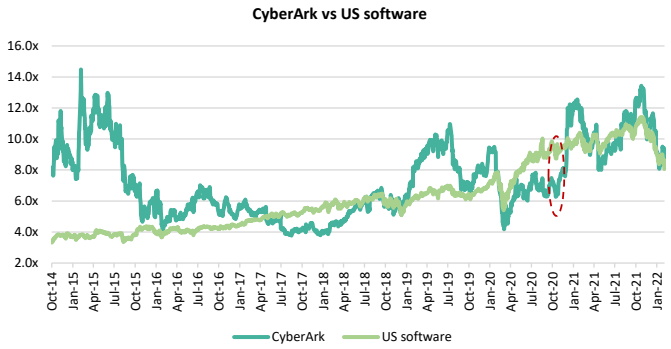
NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Chart 42 - A similar situation is observed at CyberArk

NTM EV/Revenue



Source: Jefferies, Company data, FactSet

Company Valuation/Risks

Adobe Inc.

Our $680 price target is based on 33x EV/EBITDA (slightly below median for large-cap software coverage universe), applied to our estimates in 5-8 quarters. Risks include execution at scale, competitive marketing tech market, M&A integration, and subs not being immune to recessions.

Intuit Inc.

Our $700 PT is based on a 37x EV/EBITDA (51x P/E), applied to EBITDA/EPS estimates in 5-8 quarters (i.e., 4-Q sum a year from now). Risks include slow international traction, disruptive competition, tax season volatility, law of large numbers, and COVID-19 impact.

Sage

Our 900 GBp PT is based on a 9% WACC and 3% long-term growth rate. Key risks: installed base churn, mid-market competition, renewal rate expansion, cash conversion and macro slowdown.

SAP SE

Our €130 price target is based sector FY23 FCF yield. Key risks: slower than expected S/4HANA adoption, heightened competition, cloud transition, and execution issues.

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(Article 3(1)e and Article 7 of MAR)

Recommendation Completion	February 25, 2022 , 12:29 ET.
Recommendation Distributed	February 28, 2022 , 00:00 ET.

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Jefferies Group LLC makes a market in the securities or ADRs of Intuit Inc.
Jefferies Group LLC makes a market in the securities or ADRs of SAP SE.

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Hold - Describes securities that we expect to provide a total return (price appreciation plus yield) of plus 15% or minus 10% within a 12-month period.

Underperform - Describes securities that we expect to provide a total return (price appreciation plus yield) of minus 10% or less within a 12-month period.
The expected total return (price appreciation plus yield) for Buy rated securities with an average security price consistently below $10 is 20% or more within a 12-month period as these companies are typically more volatile than the overall stock market. For Hold rated securities with an average security price consistently below $10, the expected total return (price appreciation plus yield) is plus or minus 20% within a 12-month period. For Underperform rated securities with an average security price consistently below $10, the expected total return (price appreciation plus yield) is minus 20% or less within a 12-month period.
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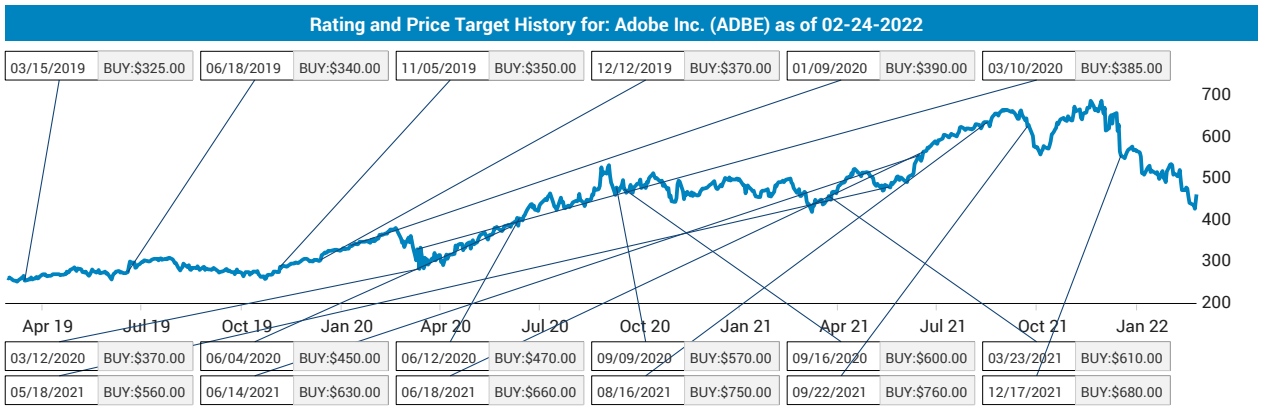
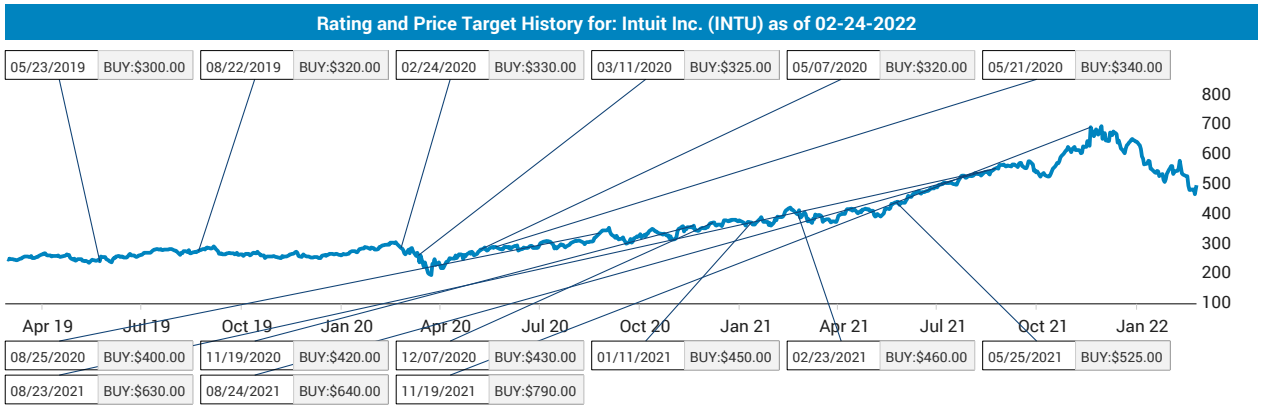
Other Companies Mentioned in This Report

- Adobe Inc. (ADBE: $465.54, BUY)
- Intuit Inc. (INTU: $487.18, BUY)
- Sage (SGE LN: p688.20, BUY)
- SAP SE (SAP: $115.02, BUY)
- SAP SE (SAP GR: €101.40, BUY)


Rating and Price Target History for: Adobe Inc. (ADBE) as of 02-24-2022
03/15/2019 BUY:$325.00
06/18/2019 BUY:$340.00
11/05/2019 BUY:$350.00
12/12/2019 BUY:$370.00
01/09/2020 BUY:$390.00
03/10/2020 BUY:$385.00
700
600
500
400
300
200
Apr 19
Jul 19
Oct 19
Jan 20
Apr 20
Jul 20
Oct 20
Jan 21
Apr 21
Jul 21
Oct 21
Jan 22
03/12/2020 BUY:$370.00
06/04/2020 BUY:$450.00
06/12/2020 BUY:$470.00
09/09/2020 BUY:$570.00
09/16/2020 BUY:$600.00
03/23/2021 BUY:$610.00
05/18/2021 BUY:$560.00
06/14/2021 BUY:$630.00
06/18/2021 BUY:$660.00
08/16/2021 BUY:$750.00
09/22/2021 BUY:$760.00
12/17/2021 BUY:$680.00


Rating and Price Target History for: Intuit Inc. (INTU) as of 02-24-2022
05/23/2019 BUY:$300.00
08/22/2019 BUY:$320.00
02/24/2020 BUY:$330.00
03/11/2020 BUY:$325.00
05/07/2020 BUY:$320.00
05/21/2020 BUY:$340.00
800
700
600
500
400
300
200
100
Apr 19
Jul 19
Oct 19
Jan 20
Apr 20
Jul 20
Oct 20
Jan 21
Apr 21
Jul 21
Oct 21
Jan 22
08/25/2020 BUY:$400.00
11/19/2020 BUY:$420.00
12/07/2020 BUY:$430.00
01/11/2021 BUY:$450.00
02/23/2021 BUY:$460.00
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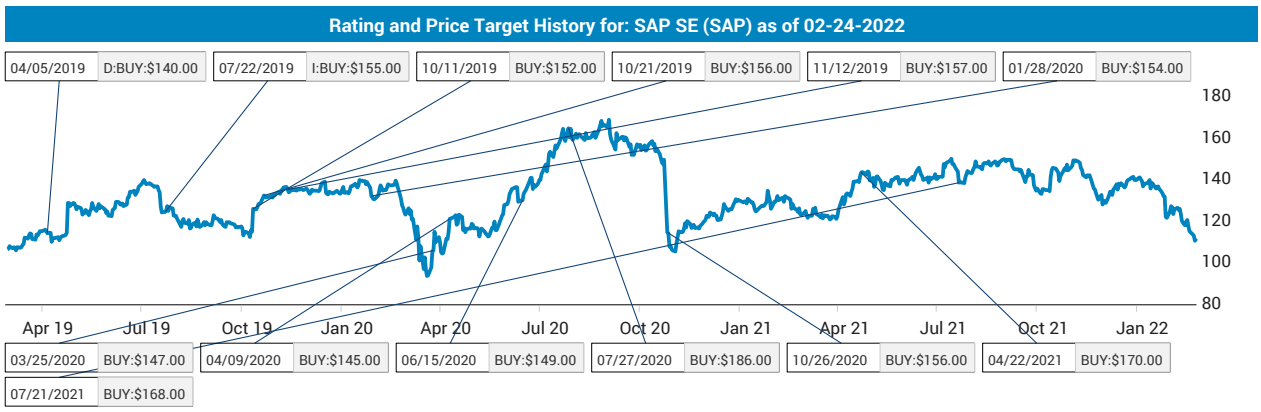
08/24/2021 BUY:$640.00
11/19/2021 BUY:$790.00


Rating and Price Target History for: SAP SE (SAP) as of 02-24-2022
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07/22/2019 I:BUY:$155.00
10/11/2019 BUY:$152.00
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01/28/2020 BUY:$154.00
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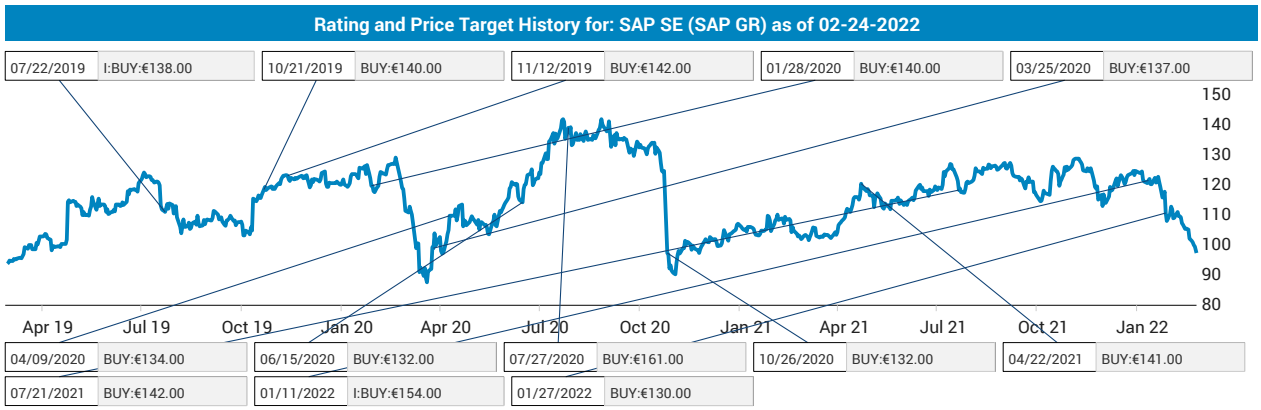

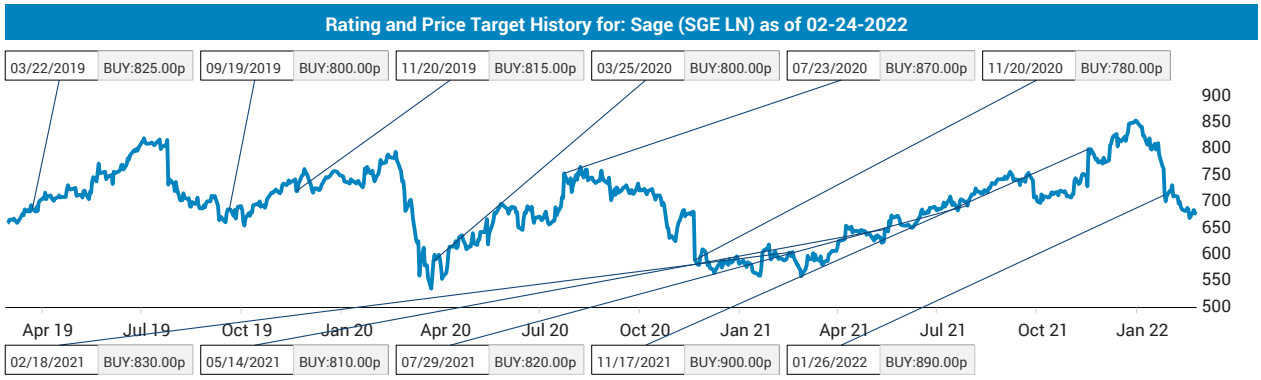

Notes: Each box in the Rating and Price Target History chart above represents actions over the past three years in which an analyst initiated on a company, made a change to a rating or price target of a company or discontinued coverage of a company.
Legend:

I: Initiating Coverage

D: Dropped Coverage

B: Buy

H: Hold

UP: Underperform

Distribution of Ratings

Distribution of Ratings						
			IB Serv./Past12 Mos.		JIL Mkt Serv./Past12 Mos.	
	Count	Percent	Count	Percent	Count	Percent
BUY	2002	64.02%	144	7.19%	26	1.30%
HOLD	997	31.88%	23	2.31%	3	0.30%
UNDERPERFORM	128	4.09%	0	0.00%	0	0.00%

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